# LINMA2710 - Scientific Computing Power Consumption

*P.-A. Absil and B. Legat*

☐ Full Width Mode ☐ Present Mode

## Table of Contents

**Energy consumption**

**Carbon intensity**

**Power consumption of computing**

**Reducing power consumption**

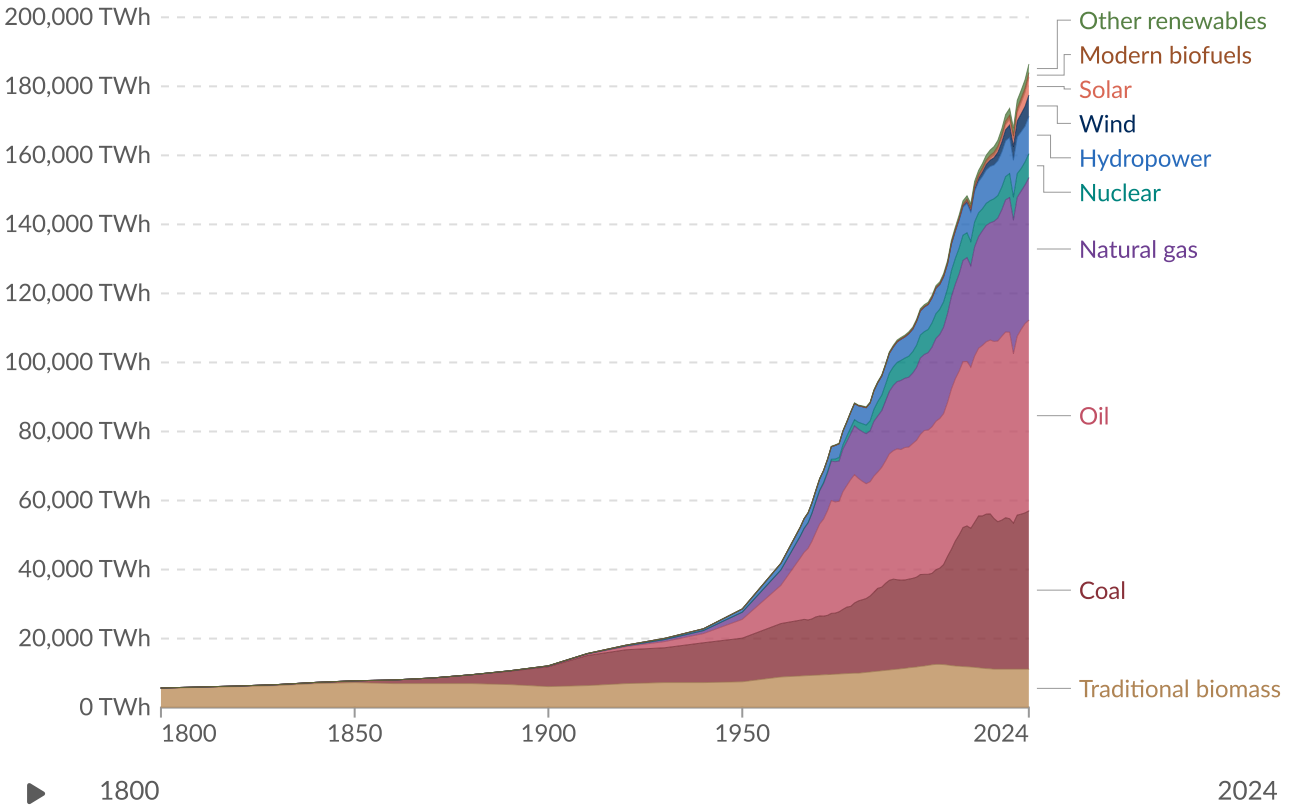
# Energy consumption

## Primary energy consumption

### Global primary energy consumption by source

Primary energy is based on the substitution method and measured in terawatt-hours.

Table Chart Settings




**Data source:** Energy Institute - Statistical Review of World Energy (2025); Smil (2017) – Learn more about this data

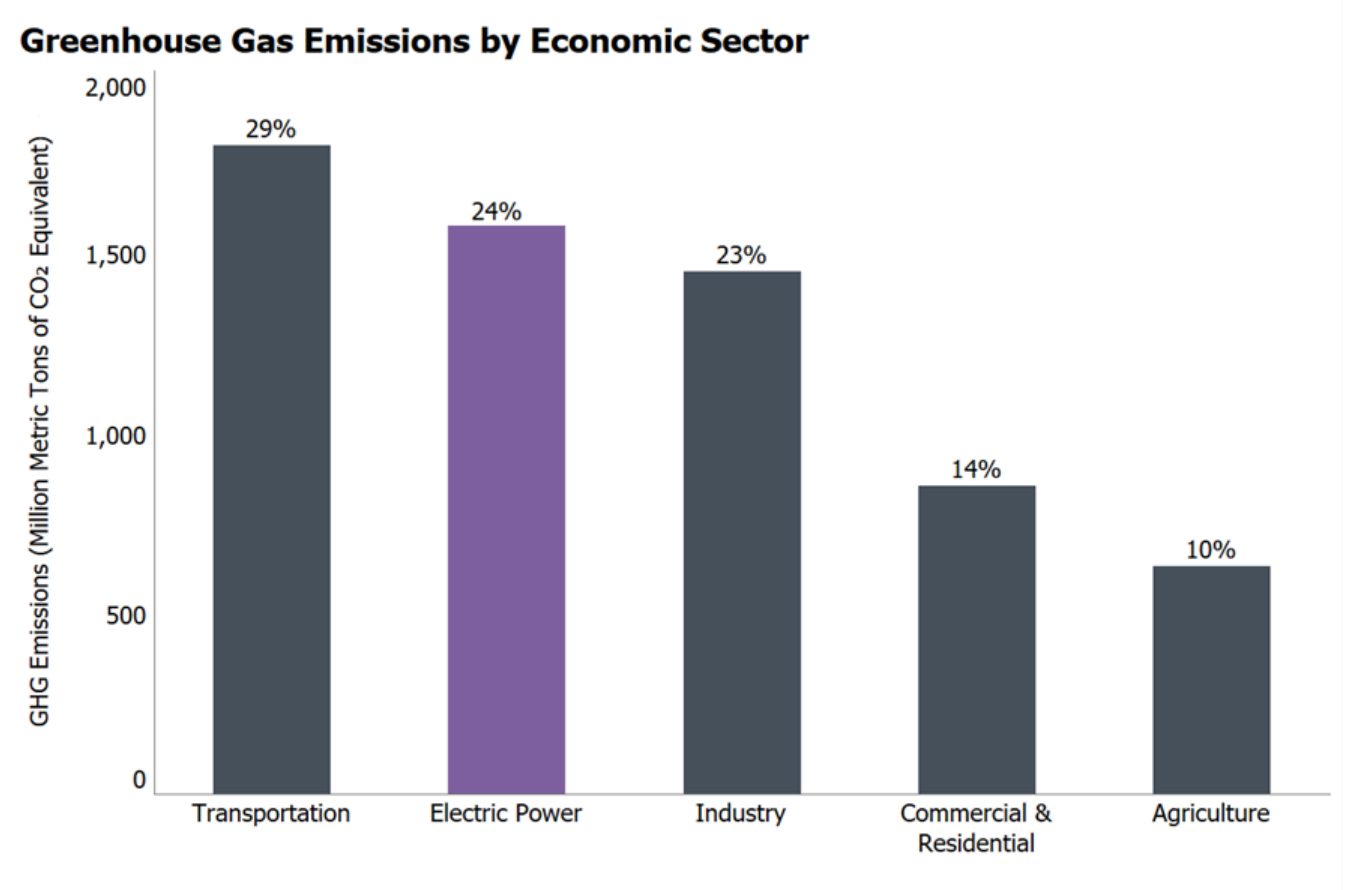
**Note:** In the absence of more recent data, traditional biomass is assumed constant since 2015.

OurWorldinData.org/energy | CC BY

Explore the data →

# The emisson from the power sector not negligible

Total U.S. Greenhouse Gas Emissions by Economic Sector in 2022



# Electricity generation per country

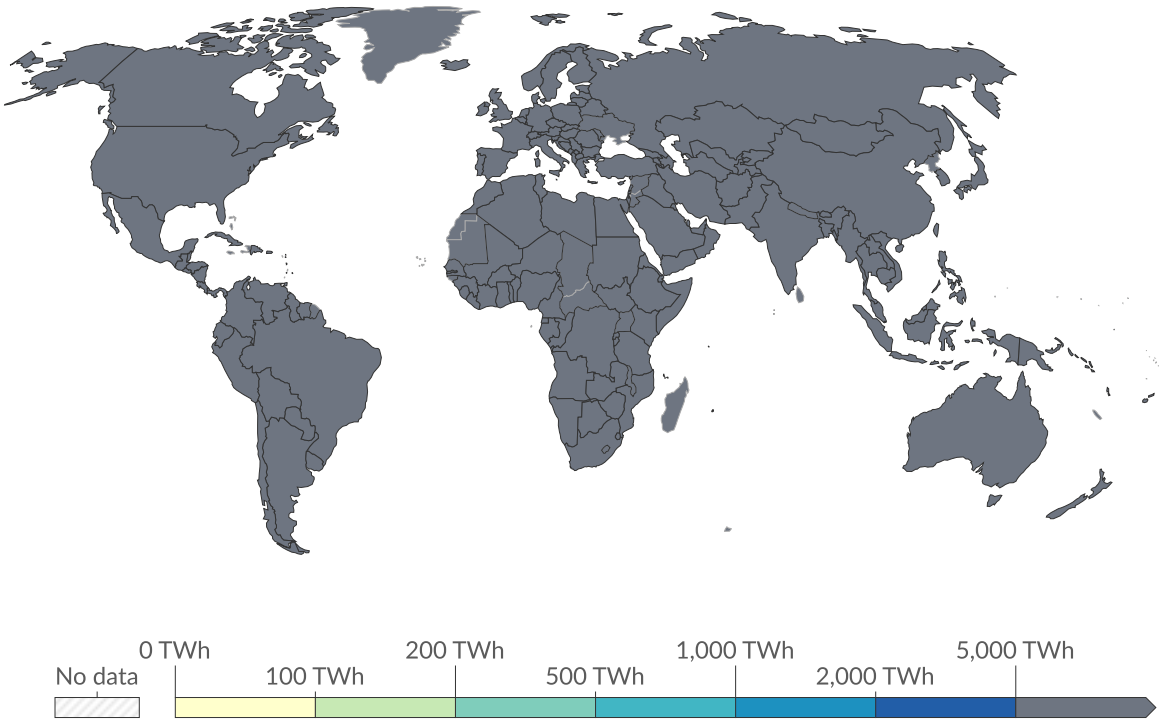
## Electricity generation, 2025

Total electricity generated in each country or region, measured in terawatt-hours.

Our World in Data

Table Map Line Bar

Select




1985 2025

**Data source:** Ember (2026); Energy Institute - Statistical Review of World Energy (2025) – Learn more about this data

OurWorldinData.org/energy | CC BY

Explore the data →

## Electricity generation per capita

# Carbon intensity

## Intensity per source

| Energy Source Carbon | Intensity (kg/MWh) |
|---|---|
| Coal | 995 |
| Petroleum | 816 |
| Natural Gas | 743 |
| Solar | 48 |
| Geothermal | 38 |
| Nuclear | 29 |
| Hydroelectricity | 26 |
| Wind | 26 |

Source

## Share of production by sources

## Share production by group

# Carbon Intensity per country

# Power consumption of computing

## RAM consumption

- 2-5 W per slot when the computer is on
- Drops to around 1 W when in sleep mode
- This is the reason your computer still consumes while sleeping.
- To turn the computer off, you need to loose all data in the RAM.
- Hibernation consists on copying the RAM to disk so that the RAM can be completely shut-off and copy it back when you turn it on.

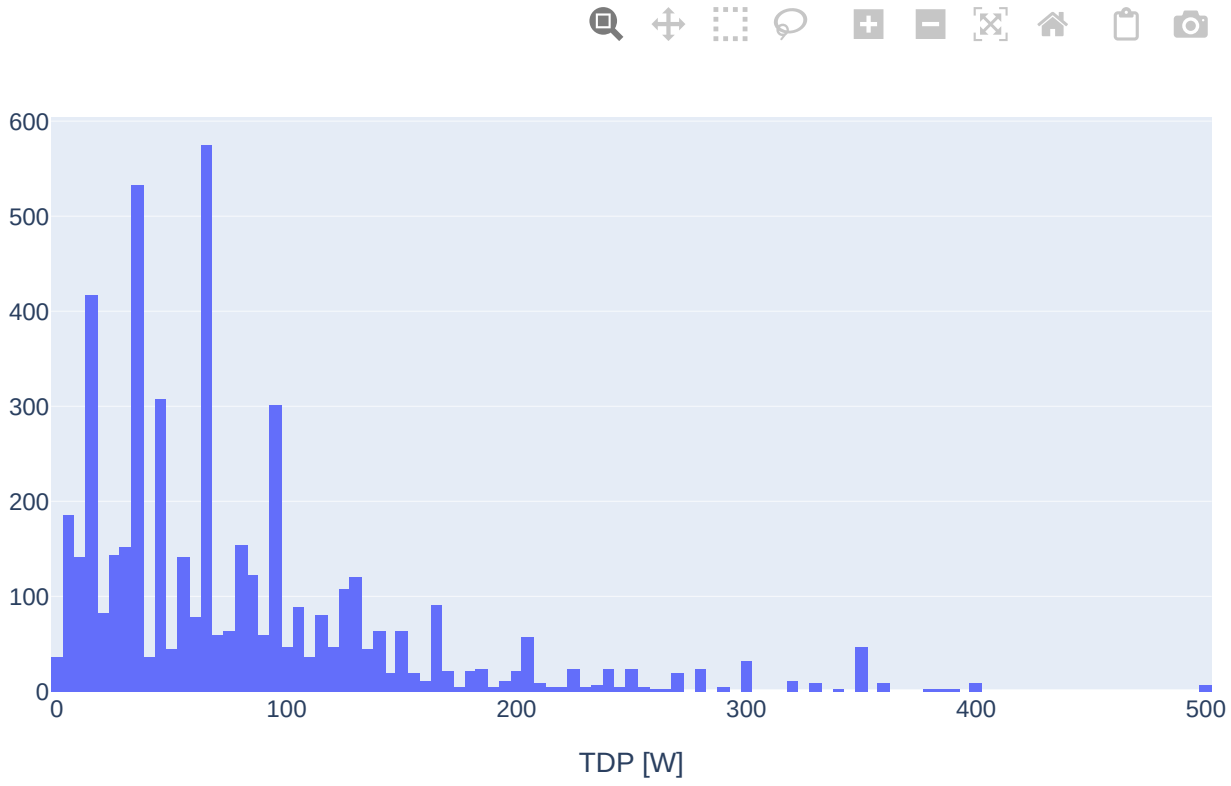
## CPU consumption : Thermal Design Power

Thermal Design Power (TDP) : Maximum amount of heat a CPU is designed to generate (important for cooling).

Power of CPU is approx. proportional to its utilization/load until below 10%

$$\texttt{power} = \texttt{TDP} \times \max(0.1, \text{load})$$

Source




| | Name | TDP |
|---|---|---|
| 1 | "AMD EPYC 9755" | 500.0 |
| 2 | "AMD EPYC 9965" | 500.0 |
| 3 | "Intel Xeon 6960P" | 500.0 |
| 4 | "Intel Xeon 6972P" | 500.0 |
| 5 | "Intel Xeon 6979P" | 500.0 |
| 6 | "Intel Xeon 6980P" | 500.0 |
| 7 | "AMD EPYC 9475F" | 400.0 |
| 8 | "AMD EPYC 9565" | 400.0 |
| 9 | "AMD EPYC 9575F" | 400.0 |
| 10 | "AMD EPYC 9655" | 400.0 |
| ⋮ more | | |
| 4898 | "Intel Quark Microcontroller D1000" | 0.0 |

## Power consumption of GPUs

# Error message

*Another cell defining gpu contains errors.*

```julia
let
    PlutoPlotly.plot(
        PlotlyBase.histogram(x=Vector(gpu[:, :tdp_watts])),
        PlotlyBase.Layout(; xaxis_title="TDP [W]")
    )
end
```

👀 Reading hidden code

# Error message

*Another cell defining gpu contains errors.*

```julia
gpu
```

👀 Reading hidden code

# Error message

```
MethodError: Cannot `convert` an object of type Int64 to an object of type String
The function `convert` exists, but no method is defined for this combination of
argument types.

Closest candidates are:
  convert(::Type{String}, ::StringManipulation.Decoration)
   @ StringManipulation
~/.julia/packages/StringManipulation/ryPnT/src/decorations.jl:365
  convert(::Type{String}, ::JSON.PtrString)
   @ JSON ~/.julia/packages/JSON/eypqd/src/lazy.jl:426
  convert(::Type{String}, ::String)
   @ Base essentials.jl:461
```

```
...
```

Show stack trace...

```julia
gpu = let
    f = Downloads.download("https://raw.githubusercontent.com/mlco2/impact/refs/heads/master/data/gpus.csv")
    df = CSV.read(f, DataFrame)
    src = "https://en.wikipedia.org/wiki/List_of_Nvidia_graphics_processing_units#Tegra_GPU"
    push!(df, ["Tesla H100-PCIE", "gpu", 350, missing, missing, missing, missing, 80, src])
    push!(df, ["Tesla H100-PCIE-80GB", "gpu", 700, missing, missing, missing, missing, 80, src])
    push!(df, ["Tesla H200-PCIE", "gpu", 600, missing, missing, missing, missing, 141, src])
    push!(df, ["Tesla H200-SXM", "gpu", 700, missing, missing, missing, missing, 141, src])
    sort!(df, :tdp_watts, rev = true)
    df
end;
```

Reading hidden code

```
⚠ thread = 1 warning: only found 8 / 9 columns around data row: 5. Filling remaining columns with `missing`
⚠ thread = 1 warning: only found 8 / 9 columns around data row: 5. Filling remaining columns with `missing`
⚠ thread = 1 warning: only found 8 / 9 columns around data row: 5. Filling remaining columns with `missing`
⚠ thread = 1 warning: only found 8 / 9 columns around data row: 5. Filling remaining columns with `missing`
⊗ Error adding value to column :memory. Maybe you forgot passing `promote=true`?
```

## Power consumption of a cluster

xAI Colossus will be the largest cluster being built (not counting mega-clusters made of several ones connected by optic fibers). Made of 100k H100 GPUs and 100k H200 GPUs so a total of 150 MW. This will need a whole new Gas turbine just to power it:

| | Source | Power | xAI need |
|---|---|---|---|
| 1 | "Nuclear plant" | 1000.0 MW | 0.15 |
| 2 | "Gas turbine" | 150.0 MW | 1.0 |
| 3 | "Wind turbine" | 3.0 MW | 50.0 |
| 4 | "1000 PV panels" | 0.32 MW | 468.8 |

Source

# Reducing power consumption

## Break down

The power consumption of a chip is the sum of two sources:

- *Static power* : primarily due to leakage currents, which become more important as the transistor size decreases.
- *Dynamic power* : Switching power given by $CV^2Af$ where
  - $C$ : Capacitance being switched
  - $V$ : Voltage
  - $A$ : *Activity factor*, i.e., number of switches of transistors per clock cycle.
  - $f$ : Clock frequency

The higher the voltage is, the higher are the leakage currents hence the power consumption but the voltage cannot be lowered without lowering the frequency hence the two are often done together → DVFS. Example of application in [YCC23].

## Dynamic voltage and frequency scaling (DVFS)

The higher the voltage is, the higher are the leakage currents hence the power consumption but the voltage cannot be lowered without lowering the frequency hence the two are often done together → DVFS. Example of application in [YCC23].

▶ **If the clock frequency is decreased, does the time performance always get worse ?**

[YCC23] J. You, J.-W. Chung and M. Chowdhury. Zeus: Understanding and Optimizing GPU Energy Consumption of DNN Training. In: 20th USENIX Symposium on Networked Systems Design and Implementation (NSDI 23) (2023); pp. 119–139.

## Gating

When a core is idle, it may first be

- *clock-gated* : part of the circuit stops switching. Corresponds to states C1-C3 in intel CPUs.

If it continues being idle, it may then

- Reduces the voltage as the clock frequency is now zero. This will reduce the leakage currents. Corresponds to state C4 in intel CPUs.

If it continues being idle, it may even be

- *power-gated* : turn off circuit blocks. Which eliminates the leakage currents. It has a fixed power cost but is worth it if it is unused for a long time [Wang et al.]. Corresponds to states C6-C10 in intel CPUs.

You can inspect the states of your cores on your laptop with

- Intel Power Gadget (Windows/macOS)
- powertop (Linux)
- HWInfo or Throttlestop (Windows)

Wang, Po-Han, Yang, Chia-Lin, Chen, Yen-Ming and Cheng, Yu-Jung *Power Gating Strategies On GPUs*, ACM Transactions On Architecture And Code Optimization (2011) 10/drfp46, cited by 32

## Reducing the power consumption of your code

▶ **DVFS and gating are automatically handled. So what actions can be taken in the design of your code or GPU kernels to allow these optimizations ?**

```
using PlutoUI, PlutoUI.ExperimentalLayout, HypertextLiteral,Luxor, BenchmarkTools,
PlutoTeachingTools, DataFrames, CSV, Downloads, Unitful
```

```
import PlutoPlotly, PlotlyBase, ShortCodes
```

```
img (generic function with 3 methods)
```